CS5525 — Assignment 3: Docker Image Demo

HyperLogistics Package Tracker CLI

Course	CS5525 - Cloud Computing
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University	University of Missouri - Kansas City
Docker Hub	mosomo82

> A terminal-based package tracking CLI built with **Python stdlib only**, containerized with Docker and published to Docker Hub under the student's own account (**mosomo82**). No third-party pip packages are used. The application runs identically locally and inside a `python:3-alpine` Docker container.

Steps & Screenshots

1 Build the Docker Image — image name contains student name

```
docker build -t tonyndemoimage .
```

The image name **tonyndemoimage** embeds the student name (Tony N).

📸 *Docker build output & `docker images` listing:*

```
PS C:\Users\mtuan\cs5525demo_repo> docker images
                                                                                                                    i Info →   U  In Use
IMAGE                    ID             DISK USAGE   CONTENT SIZE   EXTRA
tonyndemoimage:latest    9e7546ae4dce       72.7MB         17.7MB
PS C:\Users\mtuan\cs5525demo_repo> docker run tonyndemoimage
python: can't open file '/app/pranathi_16254444.py': [Errno 2] No such file or directory
PS C:\Users\mtuan\cs5525demo_repo> docker build -t tonyndemoimage .
[+] Building 1.4s (9/9) FINISHED                                                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                        0.0s
 => => transferring dockerfile: 129B                                                                                        0.0s
 => [internal] load metadata for docker.io/library/python:3-alpine                                                          0.5s
 => [auth] library/python:pull token for registry-1.docker.io                                                               0.0s
 => [internal] load .dockerignore                                                                                           0.0s
 => => transferring context: 2B                                                                                             0.0s
 => [internal] load build context                                                                                           0.0s
 => => transferring context: 172B                                                                                           0.0s
 => CACHED [1/3] FROM docker.io/library/python:3-alpine@sha256:faee120f7885a06fcc9677922331391fa690d911c020abb9e8  0.0s
 => => resolve docker.io/library/python:3-alpine@sha256:faee120f7885a06fcc9677922331391fa690d911c020abb9e8025ff3d  0.0s
 => [2/3] ADD . /app                                                                                                        0.1s
 => [3/3] WORKDIR /app                                                                                                      0.1s
 => exporting to image                                                                                                      0.5s
 => => exporting layers                                                                                                     0.2s
 => => exporting manifest sha256:2b94575f98b6aeb01977a11b5484679e2e2252698b9dc603e47af046559224a8                          0.0s
 => => exporting config sha256:5e78b8297b15ee092cc836032df88cae60808f16865094a0158c570d018819bc                            0.0s
 => => exporting attestation manifest sha256:d14c3673ea317ec09ec1b92b604d13be97547ca6a13d52261548a393f2db0175              0.0s
 => => exporting manifest list sha256:e29ac66a6b4429838e2489bad3e3291529fb268b74b6c8f0a2d0477963a7d75c                     0.0s
 => => naming to docker.io/library/tonyndemoimage:latest                                                                    0.0s
 => => unpacking to docker.io/library/tonyndemoimage:latest                                                                 0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/o7vgktnbh61ecs1ivp33ew52h
```

2 Run the Container Locally

```
docker run --rm tonyndemoimage
```

📸 *Container output — HyperLogistics banner, package table, fleet stats, route detail:*

```
PS C:\Users\mtuan\cs5525demo_repo> docker run --rm tonyndemoimage
Hello, Class.
This is a simple Python function.
PS C:\Users\mtuan\cs5525demo_repo> docker run --rm hyperlogistics
╔═══════════════════════════════════════════════════════════════════╗
│                    H Y P E R L O G I S T I C S                    │
│                                                                   │
╟───────────────────────────────────────────────────────────────────╢
│                Package Tracking System  •  v2026.3                │
│                                                                   │
│                  Real-time Shipment Intelligence                  │
│                                                                   │
╚═══════════════════════════════════════════════════════════════════╝

  SHIPMENT SUMMARY
┌──────────────┬──────────────────────────┬──────────┬──────────────┬──────────────┬──────────────┬──────────┐
│ Tracking ID  │ Description              │ Type     │ Origin       │ Destination  │ Status       │ ETA      │ kg       │
├──────────────┼──────────────────────────┼──────────┼──────────────┼──────────────┼──────────────┼──────────┼──────────┤
│ HL-2026-00101│ Industrial Sensor Array  │ Fragile  │ Seattle, WA  │ Boston, MA   │ In Transit   │ Mar 20, 2026│ 4.2      │
│ HL-2026-00247│ Medical Supply Kit       │ Priority │ Dallas, TX   │ Miami, FL    │ Out for Delivery│ Mar 17, 2026│ 1.8      │
│ HL-2026-00389│ Automotive Parts Set     │ Heavy    │ Detroit, MI  │ Phoenix, AZ  │ Processing   │ Mar 24, 2026│ 22.5     │
│ HL-2026-00412│ Consumer Electronics Bundle│ Standard│ San Jose, CA │ New York, NY │ Delivered    │ Mar 16, 2026│ 3.1      │
│ HL-2026-00558│ Pharmaceutical Shipment  │ Priority │ Atlanta, GA  │ Portland, OR │ In Transit   │ Mar 21, 2026│ 0.9      │
│ HL-2026-00634│ Lab Equipment            │ Fragile  │ Chicago, IL  │ San Diego, CA│ Processing   │ Mar 22, 2026│ 7.6      │
│ HL-2026-00721│ Textbooks & Office Supplies│ Standard│ Austin, TX   │ Charlotte, NC│ Out for Delivery│ Mar 17, 2026│ 11.3     │
│ HL-2026-00895│ Smart Home Device Pack   │ Standard │ Portland, OR │ Philadelphia, PA│ In Transit │ Mar 23, 2026│ 2.4      │
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  FLEET STATUS OVERVIEW
```

3 Log In to Docker Hub — student's own account

```
docker login
```

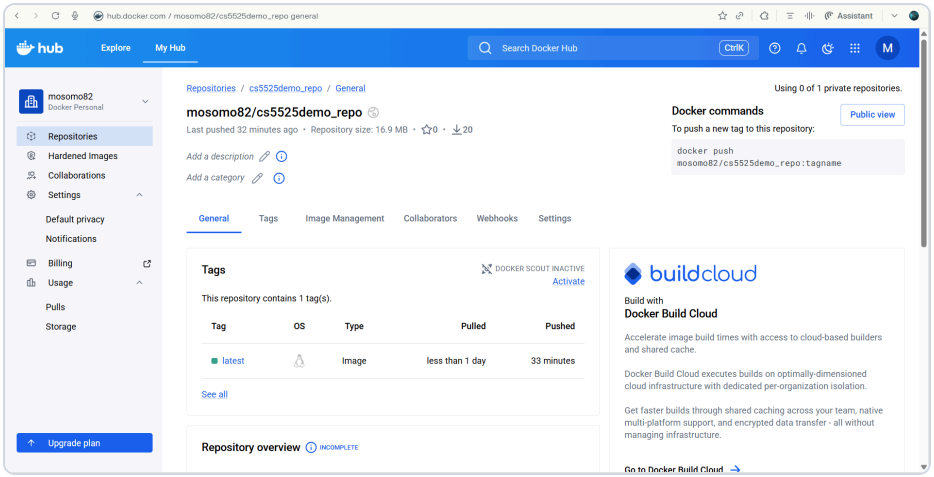
📸 *Docker Hub login success showing username **mosomo82**:*

```
PS C:\Users\mtuan\cs5525demo_repo> docker login -u "mosomo82" -p "dckr_pat_tu_jVnc5kIeOKoEinU0rfkYCaos" docker.io
WARNING! Using --password via the CLI is insecure. Use --password-stdin.
Login Succeeded
```

4 Tag the Image for Docker Hub

```
docker tag tonyndemoimage mosomo82/cs5525demo_repo
```

📸 *`docker images` showing tagged repository **mosomo82/cs5525demo_repo**:*



5 Push the Image to Docker Hub

```
docker push mosomo82/cs5525demo_repo
```

📸 *Push output confirming all layers uploaded to Docker Hub:*

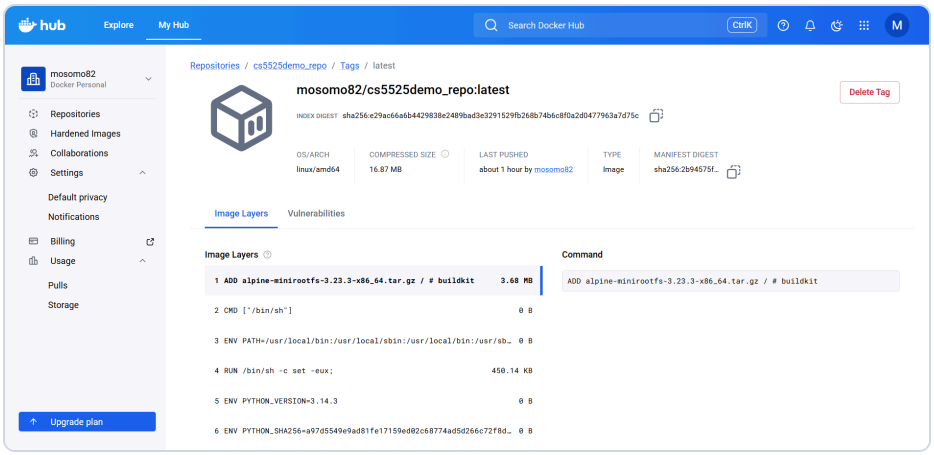
```
PS C:\Users\mtuan\cs5525demo_repo> docker push mosomo82/cs5525demo_repo
Using default tag: latest
The push refers to repository [docker.io/mosomo82/cs5525demo_repo]
f33dfdf73b5f: Pushed
950d348ff477: Pushed
4f4fb700ef54: Pushed
0805a1082be0: Pushed
2800a7aef8b1: Pushed
589002ba0eae: Pushed
3566efde290b: Pushed
latest: digest: sha256:e29ac66a6b4429838e2489bad3e3291529fb268b74b6c8f0a2d0477963a7d75c size: 856
```

6 Verify Image on Docker Hub Website

URL:
`https://hub.docker.com/repository/docker/mosomo82/cs5525demo_repo`

Confirms: account name **mosomo82**, repository public, tag **latest**.

📸 *Docker Hub web page — account name and repository visible:*



Quick Reference — All Commands

1. `docker build -t tonyndemoimage .`
2. `docker run --rm tonyndemoimage`
3. `docker login`
4. `docker tag tonyndemoimage mosomo82/cs5525demo_repo`
5. `docker push mosomo82/cs5525demo_repo`

Technical Notes

Property	Value
Base image	`python:3-alpine` — no additional packages installed
Dependencies	Python stdlib only: `datetime`, `sys`, `os`
Non-interactive safe	`sys.stdin.isatty()` gates interactive prompt — container exits cleanly
Reproducible output	8 hardcoded package records (no `random`) — identical on every run
Packages tracked	8 shipments across 4 statuses: Processing, In Transit, Out for Delivery, Delivered

CS5525 | Tony Nguyen | March 20, 2026